PRINCE2 | P2-26

Lessons Report

Lessons that should be passed to the wider organisation

Meridian Group

Programme Helios — Infrastructure and Platform Stack for AI GPU Compute at Scale

Field	Detail
Document title	Lessons Report
Document reference	*P2-26-HEL-T2-v1.0*
Project / Programme	*Helios Tranche 2 — Scale Build*
Version	*1.0*
Status	*Baselined*
Date	*18 Sep 2026*
Author	*D. Okonkwo, Programme Manager*
Owner	*Project Manager*
Approver	*Project Board; routed to the PMO for organisational action*
Classification	*Internal*
Retention	*7 years from programme closure*

Document control

Version history

Version	Date	Author	Summary of change
0.1	*04 Jun 2026*	*D. Okonkwo, Programme Manager*	*Initial draft*
0.2	*21 Jul 2026*	*D. Okonkwo, Programme Manager*	*Updated following workstream lead review*
1.0	*18 Sep 2026*	*D. Okonkwo, Programme Manager*	*Baselined at the Tranche 2 funding gate*

Reviewers

Name	Role	Review scope	Date reviewed
T. Nakamura, Chief Platform Architect	*Chief Platform Architect*	*Technical content and solution alignment*	*08 Sep 2026*
L. Haddad, CISO	*CISO*	*Security, assurance and residency content*	*10 Sep 2026*
C. Whitfield, Finance Business Partner	*Finance Business Partner*	*Cost, funding and benefit figures*	*11 Sep 2026*
M. Verity, Head of P3O	*Head of P3O*	*Governance standards and completeness*	*15 Sep 2026*

Approval

This product is baselined once approval is recorded below. Any subsequent change must go through change control.

Name	Role	Decision	Date
R. Al Mazrouei, Group CTO	*Senior Responsible Owner / Group CTO*	*Approved*	*18 Sep 2026*
H. Farrell, Director of Infrastructure	*Director of Infrastructure*	*Approved*	*18 Sep 2026*
L. Haddad, CISO	*CISO*	*Approved subject to closure of ISS-003*	*18 Sep 2026*

Related products

Reference	Product	Relationship
P2-15	Lessons Log	Source
P2-22	End Stage Report	Issued alongside
P2-23	End Project Report	Issued alongside
P3O-13	Lessons Repository	Organisational destination

How to use this template

Aspect	Guidance
Purpose	Passes on lessons that can usefully be applied to other projects, and provokes action so that positive lessons are embedded and negative ones are not repeated.
Framework	PRINCE2 — Produced at stage boundaries where warranted, and at project closure
Created when	At a stage boundary or at closure
Reviewed / updated	By the Project Board and by the PMO / centre of excellence
Owner	Project Manager
Approver	Project Board; routed to the PMO for organisational action
Format options	Document, presentation, register entry, or tool record — the content matters, not the medium.
Tailoring	Scale the depth of each section to the scale, risk and complexity of the work. Delete sections that add no value and record the tailoring decision in the governance approach.

This is a worked example. Every field has been completed with illustrative content for Programme Helios, a fictional AI GPU infrastructure and platform programme. All example content is shown in grey italic — the black text is the template itself. Replace the grey italic content with your own.

Quality criteria

Use these to check the product is fit for purpose before submitting it for approval.

- Lessons are specific enough to act on — "communicate better" is not a lesson
- Each lesson identifies who should act and what should change
- Both what worked and what did not are covered
- Statistical evidence is included where available — estimation accuracy, defect rates, change volumes

Contents

1. Report identification

Field	Detail
Project	*WS5 MLOps and developer experience*
Reporting point	[Stage boundary / Closure]
Report date	*18 Sep 2026*
Prepared by	*D. Okonkwo, Programme Manager*
Distribution	*Covers the 2,048-GPU estate and the parallel filesystem tier*

2. Executive summary

The three to five lessons that matter most. If the reader stops here, these are what they should take away.

Programme Helios consolidates fragmented AI compute onto a single shared platform in DXB-1, scaling to 2,048 GPUs in Tranche 2 and 4,096 by Tranche 3. The decision requested is release of AED 118m of Tranche 2 capital at the 18 September 2026 gate.

The return is AED 14.3m of annual public cloud rental substituted, utilisation lifted from 31% to 72%, and median training cycle time cut from 11.5 days to 3 days. The principal risk to viability is vendor GPU allocation slipping beyond November 2026 (RSK-001), which would defer the benefit by a quarter.

3. Scope of the review

Period covered, who contributed, and the method used to gather lessons.

In scope: the physical infrastructure in DXB-1 halls A and B, the compute and storage estate, the network fabric, the orchestration and scheduling platform, the MLOps and developer experience layer, the security controls required for multi-tenancy, and the operational readiness required to run the estate. Out of scope: the models themselves, business-unit data engineering, the AUH-2 second site (Tranche 3), and any change to the existing corporate WAN.

4. What worked well

Ref	Practice	Effect observed	Recommendation for reuse	Owner for adoption
WW W-01	*Applies across DXB-1 halls A and B for the Tranche 2 scale build*	*Partially met — carried into PI 5 with reduced scope*	*18 Sep 2026*	*D. Okonkwo, Programme Manager*
WW W-02	*Enforced automatically in the delivery pipeline, with evidence retained*	*Closed with no residual exposure*	*15 Oct 2026*	*K. Adeyemi, Security Architect*
WW W-03	*Maintained by the P3O as part of the assurance evidence library*	*Escalated to the Programme Board and resolved on 18 Sep 2026*	*30 Nov 2026*	*I. Petrova, Data Platform Lead*
WW W-04	*Covers the 2,048-GPU estate and the parallel filesystem tier*	*Validated — no change required to the baseline*	*15 Dec 2026*	*J. Ferreira, Network Lead*

5. What did not work well

Ref	Situation	Root cause	Effect	Recommendation	Owner for action
WDN-01	*Scoped to the three onboarding tenants: AI Research, Fraud Analytics and Retail*	*The stakeholder register recorded them as supportive on the basis of a single early conversation*	*Hall A capacity capped at 1,536 GPUs until resolved*	*30 Nov 2026*	*A. Kowalski, Data Centre Lead*
WDN-02	*Applies from hall A handover; hall B follows in Tranche 2 closure*	*Assurance was treated as a gate rather than a continuous activity*	*Training throughput approximately 9% below the business case assumption*	*15 Oct 2026*	*R. Al Mazrouei, Group CTO*
WDN-03	*Confirmed at the Tranche 1 end-of-tranche review, 18 Sep 2026*	*Congestion control tuning is specialist work that the in-house team had not done before*	*Contingency drawdown of 12% against the AED 9.4m held*	*18 Sep 2026*	*C. Whitfield, Finance Business Partner*
WDN-04	*Covers the 2,048-GPU estate and the parallel filesystem tier*	*No single forum with both parties present and authority to decide*	*On-call rota cannot be sustained beyond a 5x8 model at handover*	*25 Jun 2027*	*S. Bhatt, Project Manager*

6. Lessons by category

Category	Lesson	Recommendation	Applies to
Planning and estimating	*Hardware burn-in took 35 days against a 12-day estimate because firmware verification was performed serially on a single isolated VLAN*	*Model firmware verification as its own work package with parallel VLAN capacity sized for the shipment rate*	*Schedule development*
Governance and decision-making	*The rate card decision was deferred twice because Finance and the business units were consulted sequentially rather than together*	*Convene a single decision workshop with Finance and business unit heads together, with the sponsor present to break deadlock*	*Governance and decision-making*
Technical and design	*Engaging vendor performance engineering for two weeks during commissioning lifted all-reduce efficiency from 79% to 84% in the first pass*	*Budget two weeks of vendor performance engineering per hall as a planned activity, not as a remedial response*	*Estimating*
Security and compliance	*A critical finding on cluster API exposure was discovered late*	*Run a security design review at architecture sign-off and a first*	*Assurance planning*

Category	Lesson	Recommendation	Applies to
	because the test was scheduled after build completion	*penetration test at 30% build, not only at completion*	
Stakeholder and communication	*Fraud Analytics resistance was underestimated because engagement began after the decommissioning decision was announced*	*Assess attitude against observed behaviour, and engage those losing autonomy before the decision is communicated*	*Stakeholder engagement*
Procurement and suppliers	*Awarding fabric and platform work to a single partner created a concentration risk that later required a performance bond and step-in rights to manage*	*Include supplier concentration explicitly in source selection criteria for packages above AED 10m*	*Source selection*
Quality and testing	*Benchmarking against a representative training workload rather than a synthetic test exposed a metadata bottleneck that synthetic tests missed*	*Make workload-representative benchmarking a mandatory acceptance criterion for storage*	*Acceptance testing*
Team and resourcing	*Widening the role to remote-first within GST±3 produced four qualified candidates within three weeks after three months of local search*	*Test the necessity of location constraints before opening scarce technical roles*	*Resourcing*

7. Review of the management approaches

Approach	How well it worked	Recommendation
Risk Management Approach	*No individual risk may remain above a residual score of 12 without SRO acceptance*	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*
Quality Management Approach	*No open critical or high security findings; acceptance benchmarks met in full*	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*
Change Control Approach	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*
Communication Management Approach	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*
Benefits Management Approach	*No more than 10% variance against the benefit profile before escalation*	*Hybrid — predictive for facilities and hardware, agile for platform, MLOps and security*
Governance and Board operation	*Written pack circulated 48 hours in advance*	*31 Mar 2027*
Tailoring decisions	*Product Status Account and Configuration*	*Product Status Account and*

Approach	How well it worked	Recommendation
	Item Record replaced by the CMDB export; Checkpoint Report replaced by the ART sync record	*Configuration Item Record replaced by the CMDB export; Checkpoint Report replaced by the ART sync record*

8. Measurements and statistics

Measure	Estimated	Actual	Variance	Commentary
Effort (person-days)	*Verified at hall A acceptance on 26 Feb 2027*	*Applies across DXB-1 halls A and B for the Tranche 2 scale build*	*−6 days*	*On plan; no intervention required this period*
Duration	*10 working days*	*4 weeks*	*4 weeks*	*90 days*
Cost	*AED 118,000,000 for Tranche 2; tolerance of +5% before escalation*	*AED 118,000,000 for Tranche 2; tolerance of +5% before escalation*	*AED 118,000,000 for Tranche 2; tolerance of +5% before escalation*	*AED 118,000,000 for Tranche 2; tolerance of +5% before escalation*
Defects found post-approval	*Delegated to the Programme Manager within the agreed tolerance*	*Excluded from Tranche 2 and deferred to the AUH-2 second site*	*+3.1%*	*Amber — recovery action agreed with the workstream lead and tracked weekly*
Approved changes	*Reduce Tranche 2 GPU count from 2,048 to 1,792*	*Add coolant leak detection integration to the NOC*	*Nil*	*Dual-source 20% of Tranche 2 accelerator volume*
Risks materialised	*No individual risk may remain above a residual score of 12 without SRO acceptance*	*No individual risk may remain above a residual score of 12 without SRO acceptance*	*No individual risk may remain above a residual score of 12 without SRO acceptance*	*No individual risk may remain above a residual score of 12 without SRO acceptance*

9. Recommendations for the organisation

Route each recommendation to a body that can act — PMO, centre of excellence, procurement, architecture, security function, HR.

Ref	Recommendation	Routed to	Priority	Response
RFT-	*30 Nov 2026*	*Portfolio Board*	*High*	*Broker all vendor*

Ref	Recommendation	Routed to	Priority	Response
01				*access through the monitored in-country jump host*
RFT-02	*15 Oct 2026*	*P3O Lessons Repository*	*Medium*	*Secure written migration commitments from three business units*
RFT-03	*18 Sep 2026*	*Finance Business Partner*	*Must have*	*Confirm vendor allocation in writing before placing the structural order*
RFT-04	*25 Jun 2027*	*Head of P3O*	*Should have*	*Retune CDU set-points and model a secondary heat rejection loop*
RFT-05	*28 May 2027*	*Group Security*	*High*	*Run two live coolant-leak drills before hall A handover*
RFT-06	*23 Apr 2027*	*Portfolio Board*	*Medium*	*Enforce mutual TLS on the cluster API and retest by 22 Sep 2026*